FEDERAL SPECIFICATION

RESPIRATOR ASSEMBLIES: AIR LINE, WITH HELMET (SUPPLIED AIR)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for use of all Federal agencies.

The complete requirements for procuring the respirator described herein shall consist of this document and the issue in effect of GGG-M-125/GEN.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the specific requirements for helmet respirator assembly designed for use with a continuous flow of supplied air..

1.2 Classification.

1.2.1 Type, style, and models. Respirator assemblies shall be of the following types and models as specified (see 6.2).

Type A - Helmet assembly (general use) air line respirator.
Type B - Helmet assembly (welding use) air line respirator.

Model 1 - Respirable compressed air control unit.
Model 2 - Respirable compressed air cooling unit.
Model 3 - Respirable compressed air heating unit.
Model 4 - Respirable compressed air cooling and heating unit.

Type C - Helmet assembly (high temperature use) air line respirator.

2. APPLICABLE DOCUMENTS

Latest issue of GGG-M-125/GEN and documents reference therein, and the additional documents specified herein.

Federal Specification:

GGG-H-211 - Helmet Welders & Shield Welding Hand Held.

FSC 4240

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3. REQUIREMENTS

3.1 Approval and certification. (See GGG-M-125/GEN)

3.2 Design.

3.2.1 Type A, models 1, 2, 3, and 4 respirators. Type A, models 1, 2, 3, and 4 helmet respirator assemblies shall be of such size and design as to provide protection to the neck and head when worn by a person of average features. The helmet assembly shall be of sufficient size to allow the person to wear glasses or have a beard (see figure 1).

3.2.2 Type B, models 1, 2, 3, and 4 respirators. Type B, models 1, 2, 3, and 4 helmet respirator assemblies shall meet the requirements specified in 3.2.1. Type B face shields shall have provisions for being fitted with a welding lens meeting the requirements of GGG-H-211 (see figure 2). The welding shield shall be so designed as to completely cover the face.

3.2.3 Type C respirators. Type C helmet respirators shall be of such size and design as to provide protection to the neck and head of a person of average features, when worn for a long duration at temperatures up to 200° Fahrenheit (F). The helmet assembly shall be of sufficient size to allow the person to wear glasses or have a beard, and shall have a face shield (see figure 3).

3.3 Principal parts. Types A, and B, models 1, 2, 3, and 4, and type C helmet respirator assemblies shall consist of the following principal parts (see figures 1, 2, and 3).

(a) A positive pressure air supply system of respirable air (user supplied, not covered by this specification). (See specification sheet GGG-M-125/9.
(b) An air line hose (see 3.3.5).
(c) A quick-release coupling (see 3.3.4).
(d) Air flow or temperature unit (see 3.3.3 through 3.3.3.3).
(e) A length of breathing tube (see 3.3.2).
(f) An arrangement for attaching hose to wearer (see 3.3.6).
(g) A helmet with shroud and a face shield visor (see 3.3.1 and 3.3.1.1).

3.3.1 Helmet with shroud enclosure. Helmet respirators shall incorporate impact protective helmets with detachable shrouds (see 3.3.1.1, figures 1, 2, and 3). The helmet shall meet the impact and penetration resistance requirement specified for helmets in GGG-H-142. A transparent faceshield shall be so attached to the helmet as to make it possible to be opened during wear to provide access to the face, eyes, and mouth. The outside of type A faceshield shall be capable of being covered with a

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clear plastic sheet of suitable size to prevent scratching of the faceshield (see 3.4). The faceshield of type C helmets shall be fabricated from suitable transparent material capable of withstanding 200°F. The shroud shall form a seal with the helmet and faceshield. Types A, B, and C helmets with shrouds shall incorporate a head suspension system which is adjustable and fits all head shapes and sizes in accordance with GGG-H-142. When worn, the assembly shall accommodate the use of sunglasses, corrective eyepieces, ear plugs, or communication equipment. The helmets shall have provisions for internal air distribution to prevent annoying drafts to the head, face, and eyes and shall not generate noise levels in excess of 80 decibels absolute (dBA) in accordance with the requirements of 30 CFR, part 11.

3.3.1.1 Shroud enclosure. Shrouds shall consist of one or more layers of cloth. The shroud shall be so designed to cover the neck and the upper part of the wearer's chest and back. Shrouds for type C helmets shall be capable of reflecting heat. Shrouds shall be capable of being easily removed and replaced on the helmet.

3.3.2 Breathing tube. A length of breathing tube leading from the respiratory inlet of the hood to the control valve or orifice shall be furnished. Tubes complete with fittings shall pass the leakage tests specified in GGG-M-125/GEN.

3.3.3 Models 1, 2, 3, and 4 air control units. Models 1, 2, 3, and 4 air control units shall have a regulating valve so designed as to permit free passage of air to the wearer of the helmet. The air control unit shall be provided with a quick-release coupling and an automatic shutoff mechanism as specified in 3.3.4. All air control units shall be capable of being attached to a belt. The valve shall have a safety adjustment so arranged that accidental bumping of the valve handwheel shall result neither in a sudden flow of air nor sudden stoppage of air flow.

3.3.3.1 Model 2. In addition to the requirements of 3.3.3, the air control device of model 2 units shall be capable of lowering the incoming air temperature, (supplied to the helmet) by a minimum of 50°F.

3.3.3.2 Model 3. In addition to the requirements of 3.3.3, the air control device of model 3 units shall be capable of raising the incoming air temperature, (supplied to the helmet) to a minimum of 50°F.

3.3.3.3 Model 4. In addition to the requirements of 3.3.3, air control device of model 4 units shall be capable of lowering or raising the incoming air temperature, (supplied to the helmet) by a minimum of 40°F.

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3.3.3.4 Type C helmet respirator air control unit. The air control unit for type C helmets shall conform to the requirements of 3.3.3.1.

3.3.4 Quick-release coupling and automatic shutoff mechanism. The quick-release coupling and automatic shutoff mechanism shall provide a manually operated means for severing the air line from the mask assembly, and at the same time cutting off the air supply automatically. The quick-release coupling shall be capable of being connected and disconnected without the aid of tools and shall sustain a leak-tight union not capable of being inadvertently disconnected by normal usage. The automatic shutoff mechanism shall automatically open the air supply when the coupling is connected. The coupling shall be readily attachable to air line hose without recourse to fittings other than those supplied with the assembly.

3.3.5 Air line hose. An air line hose, with a quick-release coupling attached to one end of the hose, shall be furnished to connect the respirator's air regulating device to the air supply system. The hose assembly and quick-release coupling shall be the same type as that used on the respirator when it was approved by the BM/NIOSH. Unless otherwise specified (see 6.2), the length of hose shall be a nominal 25 feet.

3.3.6 Belt. Each type of respirator shall be furnished with a suitable belt, having a slide for ready adjustment of belt size from approximately 29 inches to 45 inches. The belt shall support the air flow control device and shall be furnished as part of the assembly.

3.4 Protective shield sheets. Unless otherwise specified (see 6.2), 200 clear protective sheets shall be furnished with each type A assembly.

3.5 Sanitization. Types A, B, and C helmet respirators shall conform to the sanitizing requirements specified in GGG-M-125/GEN.

3.6 Spare parts. When specified (see 6.2), spare parts or accessories shall be furnished.

3.7 Interchangeability.

3.8 Instructions.

3.9 Marking.

3.10 Workmanship

> See GGG-M-125/GEN.

4. QUALITY ASSURANCE PROVISIONS.

The quality assurance provisions shall conform to GGG-M-125/GEN, and the provisions specified herein.

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4.1 Examination. Helmet assemblies sampled in accordance with GGG-M-125/GEN shall be examined to verify compliance with this specification. Assemblies shall be examined for the defects listed in table I.

Table I. Classification of defects

Categories	Defects
Major	
101	Type and model not as specified.
102	BM/NIOSH approval number missing.
103	Principal parts missing.
104	Face shield does not move easily.
105	Protective shields missing or not correct quantity.
106	Spare parts missing when specified.
107	Instructions missing.
108	Belt not as specified.
109	Marking not as specified.
110	Poor workmanship.

4.2 Tests. Helmet assemblies sampled in accordance with GGG-M-125/GEN shall be subjected to the following tests:

(a) Sanitization test (See GGG-M-125/GEN).
(b) Helmet impact test (See GGG-M-125/GEN).
(c) Temperature control test (see 4.2.1).
(d) Heat test, type C helmet assembly only (see 4.2.2).

4.2.1 Temperature air control test. Models 1, 2, 3, 4, and type C helmet assemblies sampled in accordance with GGG-M-125/GEN shall be subjected to the following functional test to determine if the requirements of 3.3.3.1 through 3.3.3.3 have met:

(a) The complete respirator assembly shall be tested in a room at ambient conditions (70°F ±5°F at relative humidity 50 percent).
(b) A suitable temperature indicator or sensor shall be located inside the helmet to record the temperature within the helmet.
(c) Following the instructions of the manufacturer, the air temperature control unit shall be operated to determine if compliance with the respective requirements of 3.3.3.1, 3.3.3.2, and 3.3.3.3 have been met. Failure to meet the requirements shall be cause for rejection.

4.4.2 Heat test for type C respirators. One type C respirator sampled from each lot in accordance with GGG-M-125/GEN shall be subjected

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for 1 hour to 200°F ±5°F in a circulating air oven. After 1 hour, the assembly shall be removed from the oven, cooled to room temperature, and examined for signs of crazing, cracking, or distortion by any of the components. Any defects shall be cause for rejection.

5. PREPARATION FOR DELIVERY. (See GGG-M-125/GEN)

6. NOTES

6.1 Intended use. Respirators covered in the specification are intended to protect the respiratory organs of the wearer from toxic vapors, dusts, metal fumes, and mists in concentrations that are not immediately dangerous to life and to protect the wearer's head from being hit by objects. Respirators covered by this specification require positive pressure inside the facepiece in relation to the immediate environment during inhalation and exhalation.

6.2 Ordering data. Purchasers should select the preferred options offered herein and include the following data in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and model of respirator required (see 1.2.1 and 3.2).
(c) If length of air line hose is different (see 3.3.5).
(d) If number of protective shields required are different (see 3.4).
(e) If spare parts are required (see 3.6).
(f) If lot size is different (see 4.2 of GGG-M-125/GEN).
(g) Level of packaging and level of packing required (see 5.1 and 5.2 of GGG-M-125/GEN.

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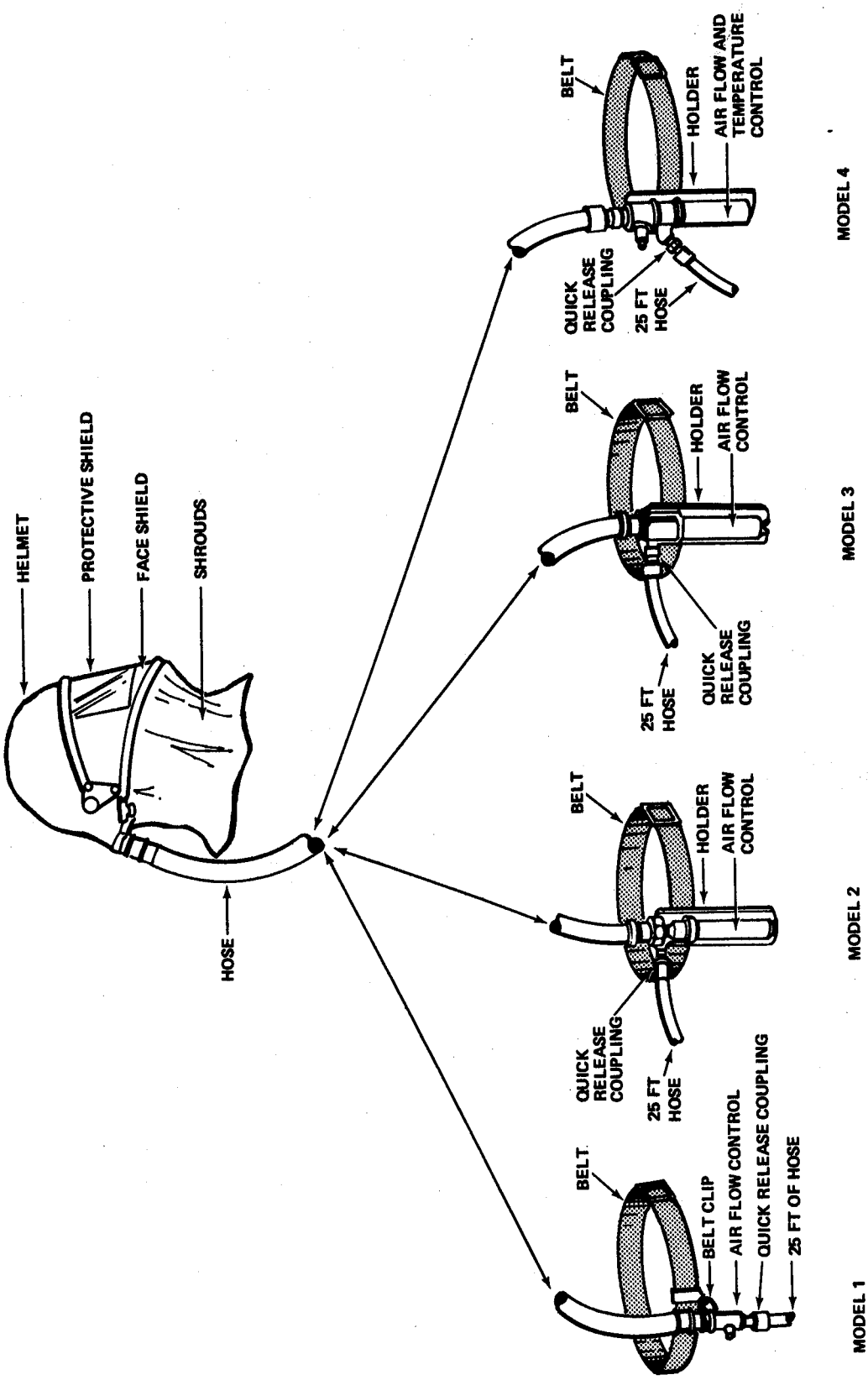


Figure 1. Type A, helmet assembly (general use) airline respirator.
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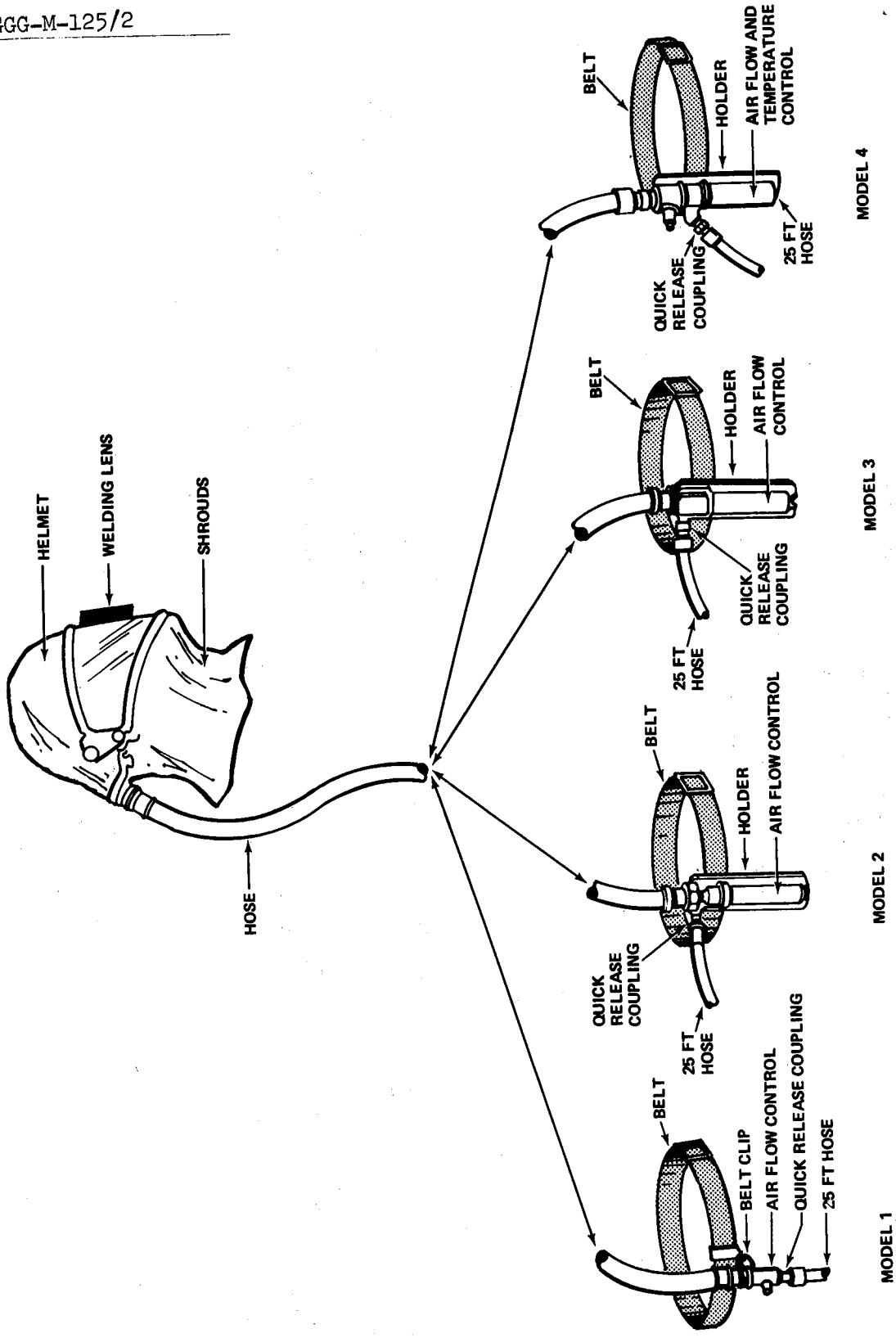


Figure 2. Type B, helmet assembly (welding use) airline respirator. (For illustrative purposes only.)

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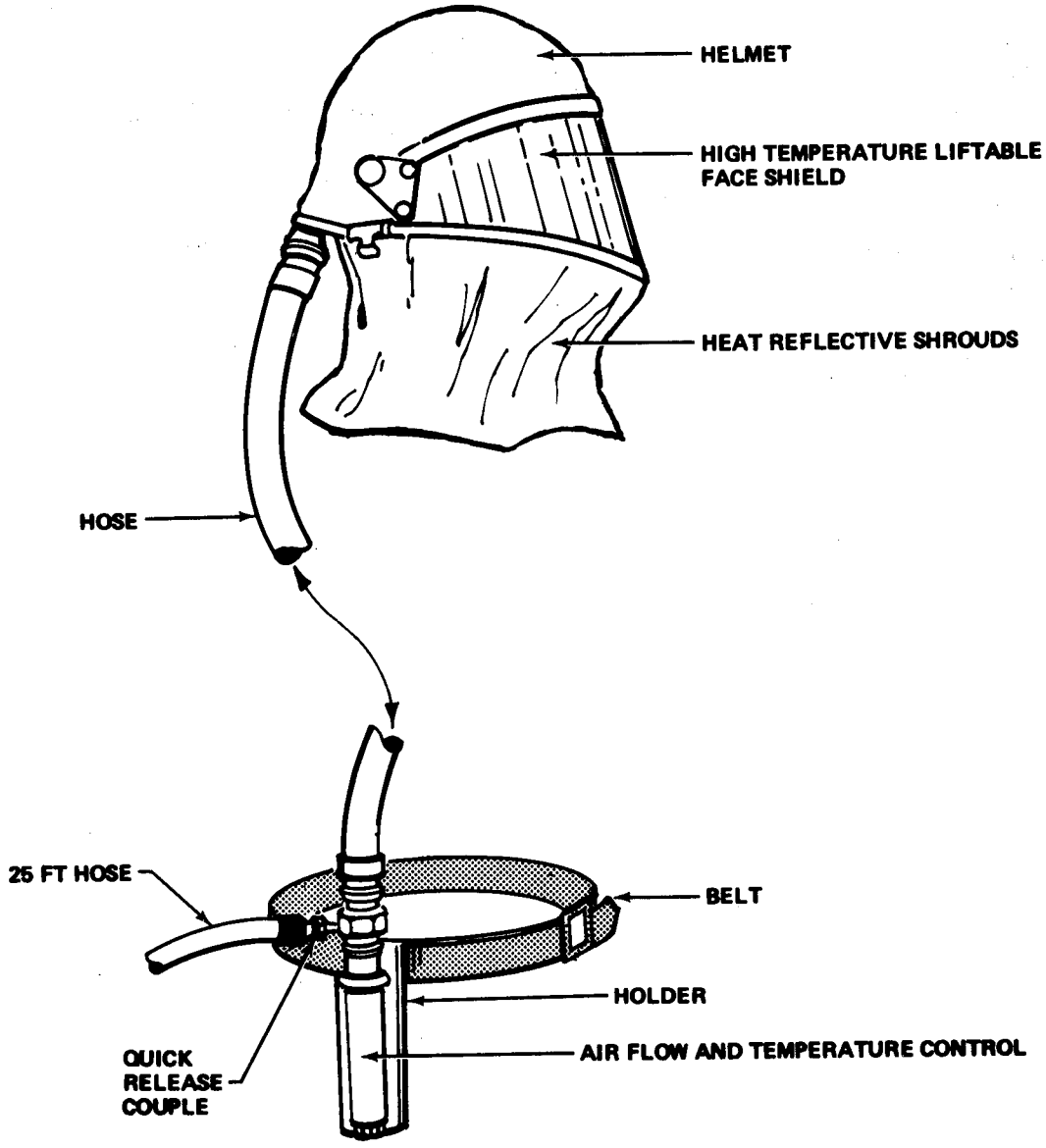


Figure 3. Type C, helmet assembly (high temperature use) airline respirator.
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